

TASK-1

FILE HANDLING AND AUTOMATION USING PYTHON

Internship Task Report

Submitted By

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1. Objective

The objective of this task is to understand Python file handling, automation logic, and exception handling. The project demonstrates how to create, read, rename, move, and delete files using Python with proper error handling techniques.

2. Tools and Technologies Used

- Python
- Visual Studio Code
- os Module
- shutil Module

3. Requirements Covered

The following requirements were successfully implemented:

- ✓ Read and write text files
 - ✓ Automate file operations
 - ✓ Rename files
 - ✓ Move files between folders
 - ✓ Delete files
 - ✓ Use try-except exception handling
 - ✓ Add comments for code explanation
-

4. Project Folder Structure

file_handling_task/

|

├── automation.py

├── sample.txt

└── backup/

(Add screenshot of folder structure here)

5. Code Explanation

Step 1: Import Required Modules

The os module is used for file operations like renaming and deleting files.

The shutil module is used for moving files between folders.

(Add code screenshot)

Step 2: File Write Operation

A text file named sample.txt is created and data is written into it using write mode.

(Add screenshot of code and output)

Step 3: File Read Operation

The contents of the text file are read and displayed in the terminal.

(Add screenshot)

Step 4: File Rename Operation

The file sample.txt is renamed to renamed_sample.txt using os.rename().

(Add screenshot)

Step 5: File Move Operation

The renamed file is moved into the backup folder using shutil.move().

(Add screenshot)

Step 6: File Delete Operation

The file inside the backup folder is deleted using `os.remove()`.

(Add screenshot)

6. Exception Handling

Exception handling is implemented using try-except blocks to prevent the program from crashing when errors occur.

Examples:

- File not found errors
- Invalid file operations
- Permission issues

(Add screenshot of exception handling code)

7. Complete Python Code

(Paste complete automation.py code here)

8. Output

Sample terminal output:

File created and data written successfully.

File Content:

Hello, this is a file handling internship task.

Python makes automation easy.

File renamed successfully.

File moved to backup folder successfully.

File deleted successfully.

(Add terminal output screenshot)

9. Insights / Findings

1. Python makes file handling simple and efficient.
 2. Automation reduces manual file management work.
 3. Exception handling improves program reliability.
 4. Built-in modules like `os` and `shutil` simplify automation tasks.
 5. Proper comments improve code readability and maintenance.
-

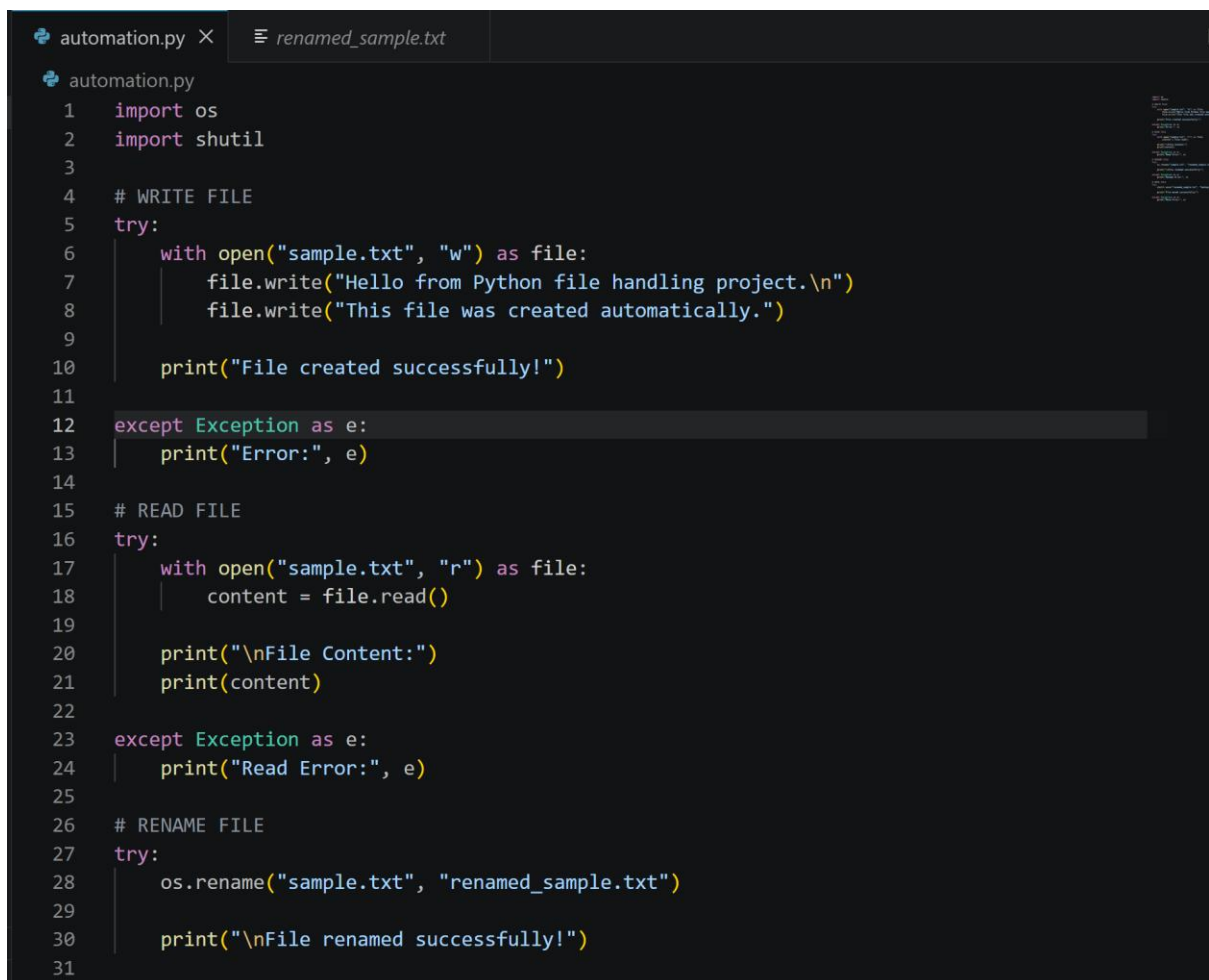
10. Challenges Faced

- Understanding file paths
 - Managing folder structure
 - Handling exceptions properly
-

11. Conclusion

This project helped in understanding Python file handling and automation concepts. Different file operations such as creating, reading, renaming, moving, and deleting files were successfully implemented along with proper exception handling techniques.

12. Screenshots

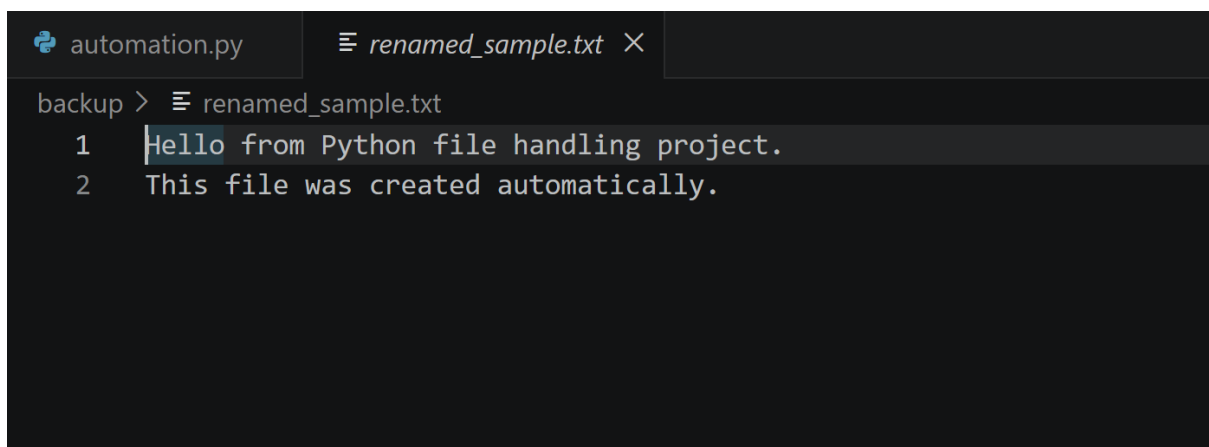


```
automation.py X  renamed_sample.txt
automation.py
1  import os
2  import shutil
3
4  # WRITE FILE
5  try:
6      with open("sample.txt", "w") as file:
7          file.write("Hello from Python file handling project.\n")
8          file.write("This file was created automatically.")
9
10     print("File created successfully!")
11
12 except Exception as e:
13     print("Error:", e)
14
15 # READ FILE
16 try:
17     with open("sample.txt", "r") as file:
18         content = file.read()
19
20     print("\nFile Content:")
21     print(content)
22
23 except Exception as e:
24     print("Read Error:", e)
25
26 # RENAME FILE
27 try:
28     os.rename("sample.txt", "renamed_sample.txt")
29
30     print("\nFile renamed successfully!")
31
```

```

31
32     except Exception as e:
33         print("Rename Error:", e)
34
35     # MOVE FILE
36     try:
37         shutil.move("renamed_sample.txt", "backup/renamed_sample.txt")
38
39         print("File moved successfully!")
40
41     except Exception as e:
42         print("Move Error:", e)

```



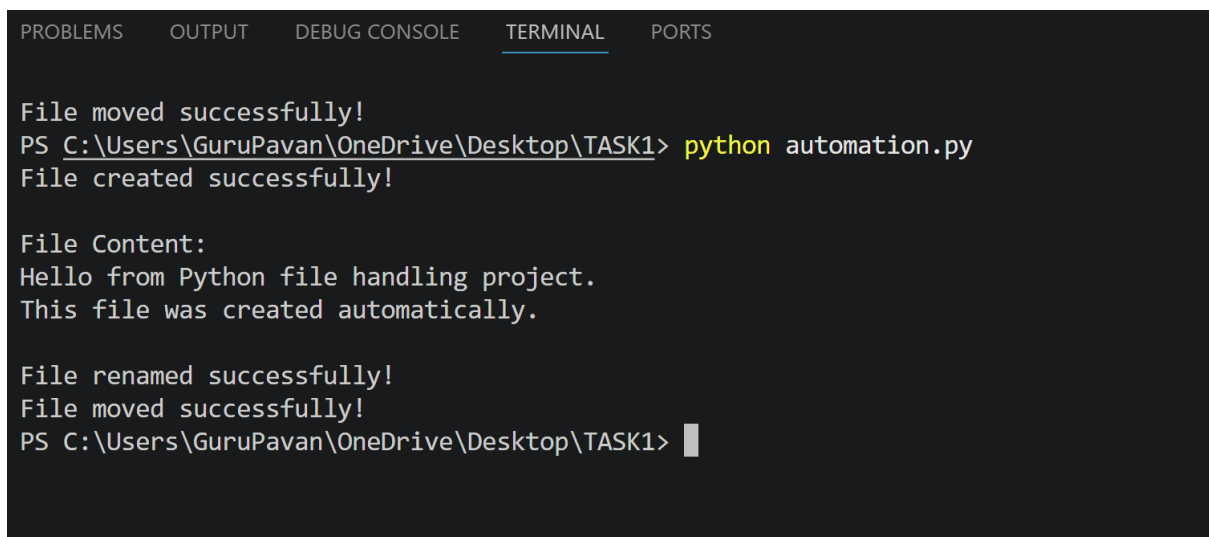
The screenshot shows a code editor with two tabs: 'automation.py' and 'renamed_sample.txt'. The 'renamed_sample.txt' tab is active, showing the following content:

```

backup > renamed_sample.txt
1 Hello from Python file handling project.
2 This file was created automatically.

```

OUTPUT:



The screenshot shows a terminal window with the following output:

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

File moved successfully!
PS C:\Users\GuruPavan\OneDrive\Desktop\TASK1> python automation.py
File created successfully!

File Content:
Hello from Python file handling project.
This file was created automatically.

File renamed successfully!
File moved successfully!
PS C:\Users\GuruPavan\OneDrive\Desktop\TASK1>

```

13. GitHub Repository Link

<https://github.com/tsgurupavankumar-creator/Alfido-python-developer/tree/258d6731aabb75abf07c46e48037e78fe56ba179/TASK1>